SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

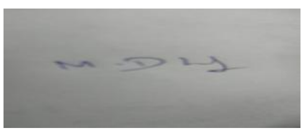
Dave's Taxonomy: Articulation (Level 4) **Total Marks: 40**
Question Count: 5
Duration :60 Minutes

Code of Conduct

I DHILIBAN/192421166 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Industry Problem Solving Task

1. **Retail Data Optimization Problem -** A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

2. **Social Media Fraud Detection Problem -** A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

4. **Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. **Agriculture Smart Farming Problem**
A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

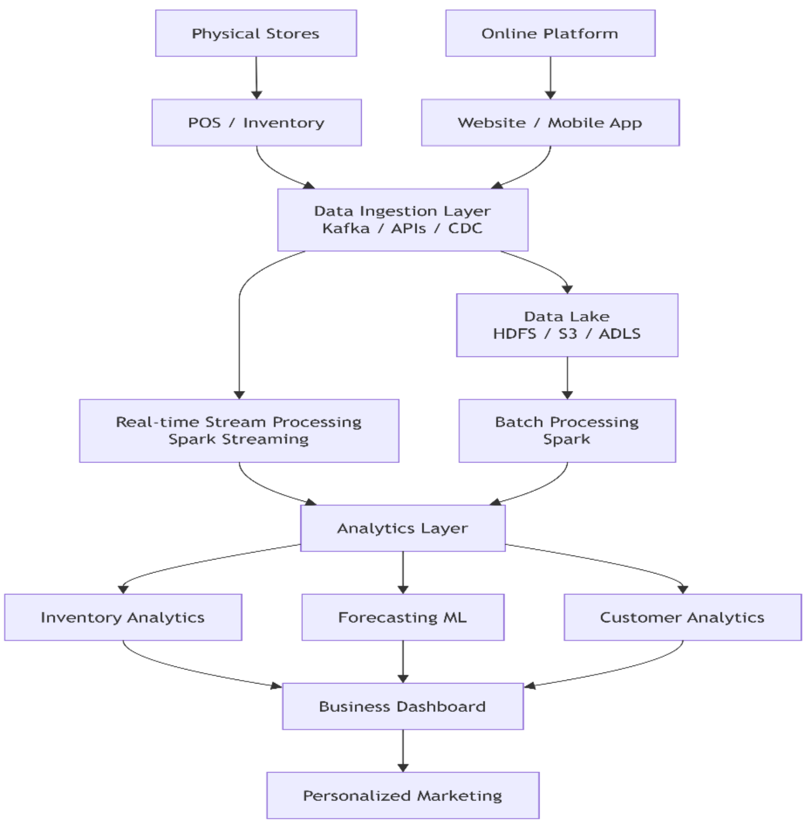
Total Marks Awarded:

1. Retail Data Optimization Problem

Problem Understanding

The retail company has both physical stores and an online platform. It generates sales, inventory, customer browsing, purchase, and feedback data. The main requirements are:

- Real-time inventory tracking
- Demand forecasting
- Personalized marketing
- Integration of online and offline data
- Scalable and secure data processing



Tools and Technologies

1. Apache Kafka: Collects real-time sales, inventory, and customer activity data.

2. Data Lake: HDFS or cloud storage such as Amazon S3 stores large historical datasets.

3. Apache Spark: Performs distributed batch and real-time processing.

4. NoSQL Database: Cassandra or MongoDB can store rapidly changing inventory and customer information.

5. Data Warehouse: Stores structured data for business intelligence and reporting.

Analytics Models

Demand Forecasting: Time-series models such as ARIMA, Prophet, or machine learning models such as XGBoost can predict future product demand.

Recommendation System: Collaborative filtering and content-based filtering can recommend products based on customer behavior.

Customer Segmentation: K-Means clustering can group customers according to purchasing behavior.

Example

If the system notices that a particular product is selling rapidly in Chennai stores, real-time analytics can update inventory levels and the forecasting model can predict increased future demand. The company can automatically reorder stock and send personalized offers to customers interested in that product.

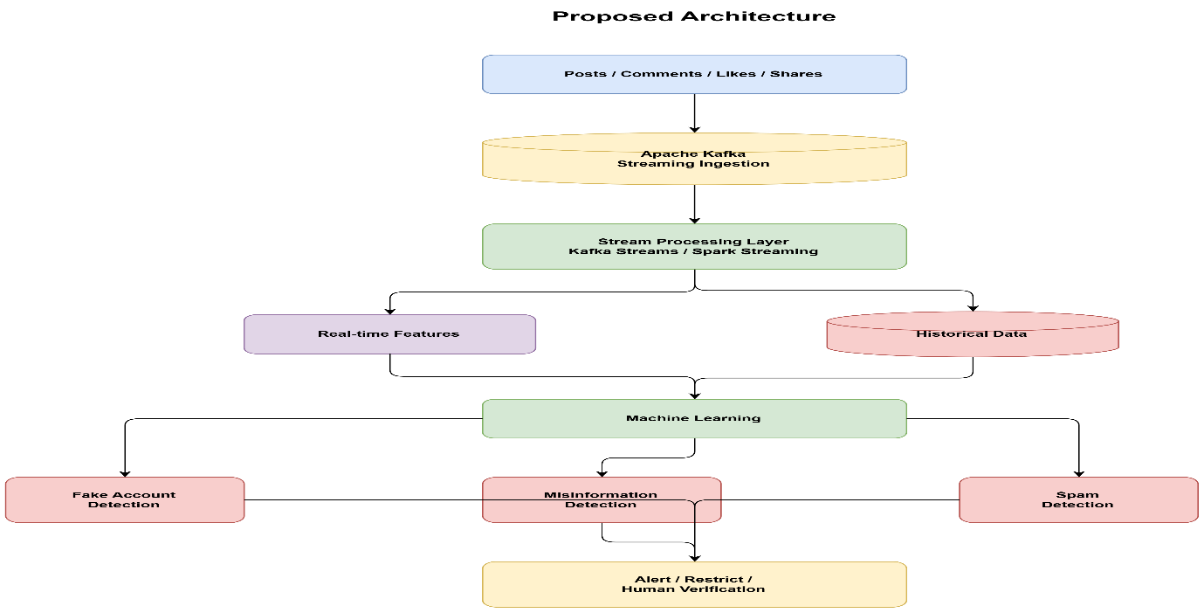
Benefits

- Real-time inventory visibility
- Reduced stock-outs and overstocking
- Better demand prediction
- Personalized customer experience
- Increased sales and customer loyalty

2. Social Media Fraud Detection Problem

Problem Understanding

A social media platform processes billions of posts, comments, likes, and shares every day. Fake accounts and misinformation must be detected quickly while handling extremely high data volume.



Stream Processing

Apache Kafka receives billions of events continuously.

Spark Streaming or **Kafka Streams** processes events in near real time.

Features can include:

- Account age
- Posting frequency
- Number of followers
- Like/share patterns
- IP/device behavior
- Repeated content
- Network connections
- Sudden abnormal activity

Machine Learning

For fake-account detection, algorithms such as:

- Random Forest
- XGBoost
- Logistic Regression
- Neural Networks

can classify accounts as legitimate or suspicious.

For misinformation, Natural Language Processing (NLP) can analyze text, while image/video analysis models can examine multimedia content.

Graph analytics can also identify suspicious networks of interconnected fake accounts.

Example

If a newly created account follows thousands of users, posts hundreds of messages within a short period, and repeatedly shares identical content, the system can assign a high-risk score and trigger additional verification.

Scalability

Kafka partitions allow data to be processed across multiple nodes. Spark distributes processing across a cluster. Cloud-based infrastructure can automatically scale when interaction volume increases.

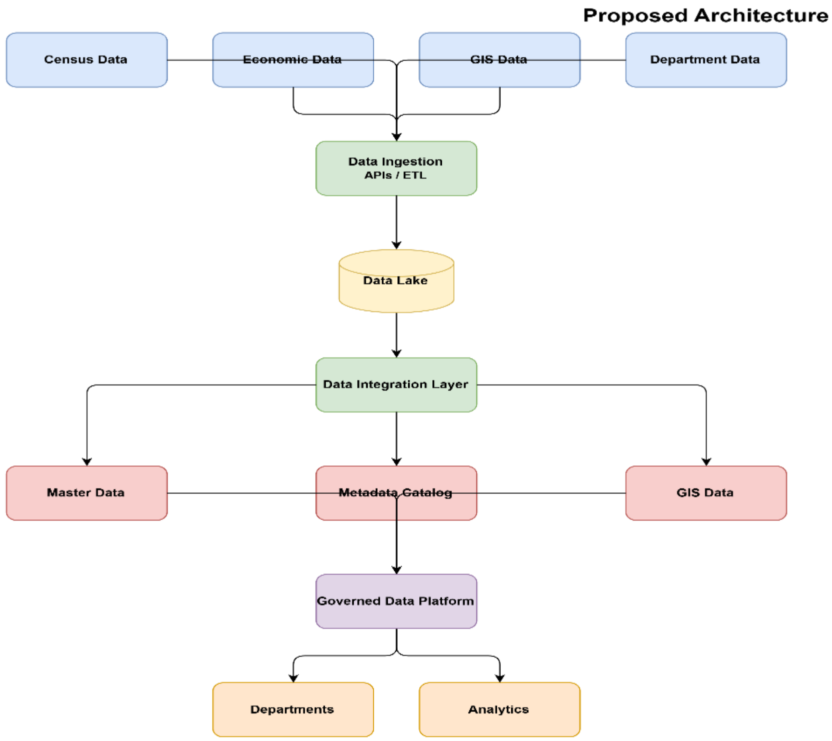
Benefits

- Near-real-time detection
- Handles billions of events
- Scalable processing
- Automated identification of suspicious behavior
- Reduced platform abuse

3. Government Data Platform Problem

Problem Understanding

The government platform combines census, economic, and geospatial information from multiple departments. The major challenges are interoperability, data consistency, privacy, security, and controlled data sharing.



Data Governance Strategy

1. Data Standards: Establish common formats, naming conventions, identifiers, and schemas across departments.

2. Metadata Management: Maintain information about the source, owner, format, quality, and usage of every dataset.

3. Master Data Management: Maintain consistent master information such as geographic areas, departments, and administrative units.

4. Data Quality: Perform validation, deduplication, consistency checks, and missing-value detection.

5. Data Catalog: A centralized catalog allows departments to discover available datasets.

Security and Privacy

- Role-Based Access Control (RBAC)
- Encryption at rest and in transit
- Multi-factor authentication
- Data masking and anonymization
- Audit logs
- Data classification
- Regular security assessments

- Controlled API access

Sensitive personal information should only be accessible to authorized personnel.

Interoperability

APIs and common data standards allow different government departments to exchange information. A common schema and unique identifiers prevent duplicate or incompatible records.

Example

The census department can provide population data while the economic department provides employment statistics. Integrating both datasets allows policymakers to identify regions with high unemployment and plan targeted development programs.

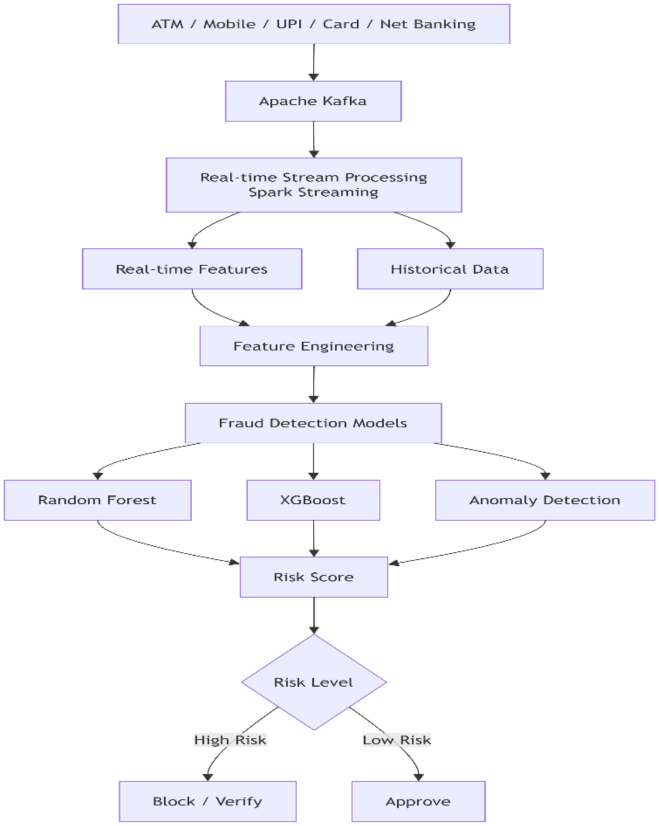
Benefits

- Single source of reliable information
- Better interdepartmental collaboration
- Improved policy decisions
- Strong privacy protection
- Standardized data exchange
- Reduced data duplication

4. Banking Fraud Detection Problem

Problem Understanding

Banks process millions of transactions every minute. Fraud detection must operate with **very low latency**, high accuracy, and the ability to analyze both historical and real-time transaction data.



Technologies

Apache Kafka: Receives transaction events continuously.

Apache Spark Streaming: Processes transactions with low latency.

Data Lake: Stores historical transactions for model training.

NoSQL Database: Provides fast access to customer and transaction features.

Machine Learning: Detects patterns associated with fraud.

Algorithms

Random Forest: Useful for classification and robust against different transaction features.

XGBoost: Provides high predictive performance for structured transaction data.

Anomaly Detection: Identifies transactions that differ significantly from normal customer behavior.

Features

The model can analyze:

- Transaction amount
- Time

- Location
- Device
- Merchant
- Transaction frequency
- Previous transaction patterns
- Customer spending behavior

Example

A customer normally spends ₹1,000–₹3,000 per transaction in Chennai. Suddenly, several large transactions occur from a new device and distant location within minutes. The system can calculate a high fraud probability and request additional authentication.

Security and Risk Controls

- Encryption
- Tokenization
- Access control
- Audit logs
- Multi-factor authentication
- Continuous model monitoring
- Human review for high-risk transactions

Benefits

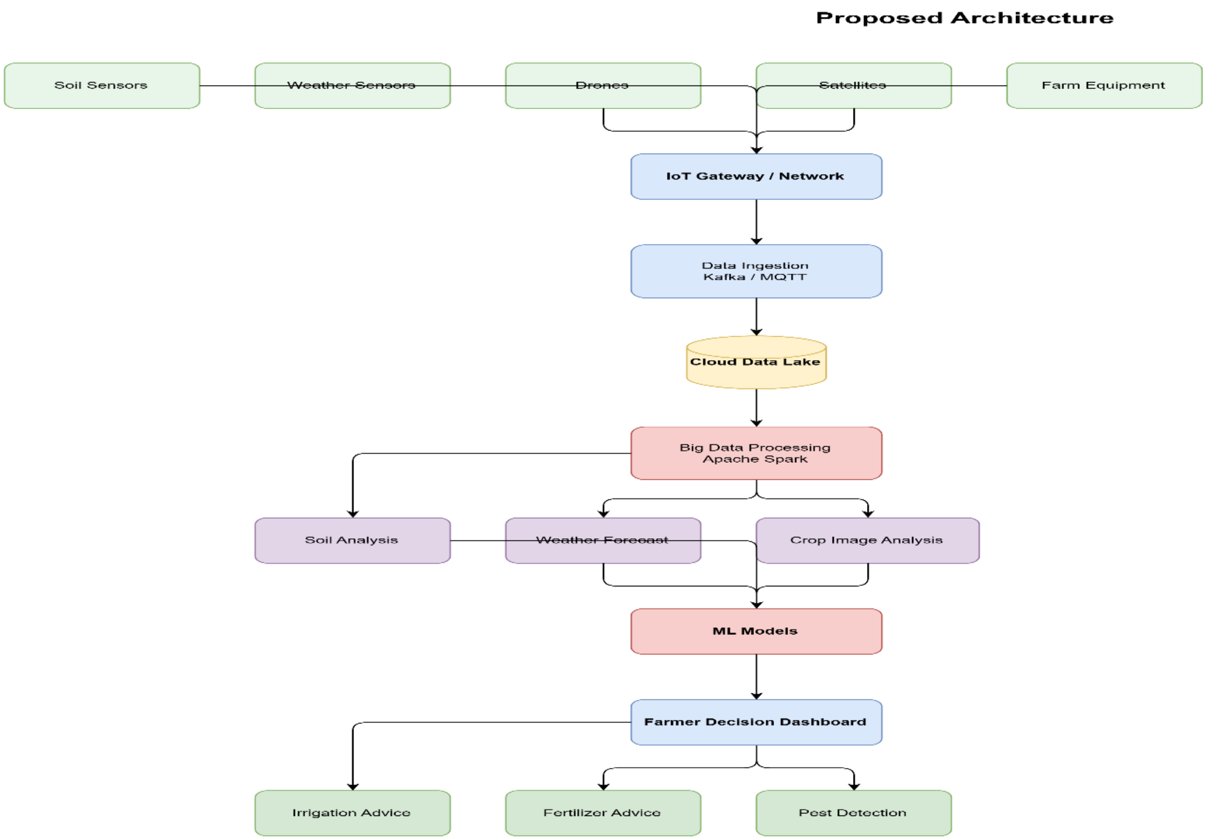
- Real-time fraud detection
- Low response latency
- Reduced financial losses
- Improved fraud prediction
- Scalable transaction processing

A combination of streaming analytics and machine learning is more effective than relying only on offline fraud analysis.

5. Agriculture Smart Farming Problem

Problem Understanding

Smart farming collects information from soil sensors, weather stations, drones, satellites, and crop monitoring systems. The data is distributed and continuously generated. Big data analytics can convert this information into useful recommendations for farmers.



Technologies

IoT Sensors: Collect soil moisture, temperature, humidity, pH, and other field information.

MQTT/Kafka: Transmit sensor data continuously.

Cloud Data Lake: Stores large volumes of sensor, drone, weather, and crop data.

Apache Spark: Processes distributed agricultural datasets.

GIS: Helps analyze geographical differences between fields.

Machine Learning: Predicts crop yield, irrigation requirements, and disease risks.

Analytics Models

1. Yield Prediction: Regression and machine learning models can estimate crop yield using soil, weather, and historical crop data.

2. Irrigation Prediction: Soil moisture and weather data can determine when irrigation is required.

3. Disease Detection: Computer vision models can identify diseases from drone or smartphone crop images.

4. Weather Analytics: Historical and real-time weather information can help farmers plan planting and harvesting.

5. Resource Optimization: Analytics can determine the required amount of water, fertilizer, and pesticides.

Example

If soil sensors detect low moisture while weather data predicts no rainfall, the system can recommend irrigation for that particular field area instead of watering the entire farm.

Benefits

- Increased crop yield
- Reduced water consumption
- Optimized fertilizer usage
- Early disease detection
- Reduced farming costs
- Better resource utilization
- Data-driven farming decisions

Security and Risk Awareness

Agricultural data should be protected using authentication, encryption, access control, and secure IoT communication. Sensor failures and inaccurate readings should also be detected through data validation.